# MPG No. 15.4.56 $4'/mm'm$ (4'/mm'm setting) [ Type III, tetragonal ]

Table 1: Wyckoff site: `1o`, site symmetry: `4'/mm'm`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, 0] | `[1,2,3,4,5,6,7,8,9,10,11,12,13,14,15,16]` |

Table 2: Wyckoff site: `2a`, site symmetry: `4'm'm`

| No. | position | mapping |
|---|---|---|
| 1 | [0, 0, $z$] | `[1,2,7,8,11,12,13,14]` |
| 2 | [0, 0, $-z$] | `[3,4,5,6,9,10,15,16]` |

Table 3: Wyckoff site: `4b`, site symmetry: `m.m2`

| No. | position | mapping |
|---|---|---|
| 1 | [$x$, $x$, 0] | `[1,4,6,7]` |
| 2 | [$-x$, $-x$, 0] | `[2,3,5,8]` |
| 3 | [$-x$, $x$, 0] | `[10,11,13,16]` |
| 4 | [$x$, $-x$, 0] | `[9,12,14,15]` |

Table 4: Wyckoff site: `4c`, site symmetry: `m2'm.`

| No. | position | mapping |
|---|---|---|
| 1 | [$x$, 0, 0] | `[1,6,9,14]` |
| 2 | [$-x$, 0, 0] | `[2,5,10,13]` |
| 3 | [0, $x$, 0] | `[4,7,11,16]` |
| 4 | [0, $-x$, 0] | `[3,8,12,15]` |

Table 5: Wyckoff site: `8d`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | [$x$, $y$, 0] | `[1,6]` |
| 2 | [$-x$, $-y$, 0] | `[2,5]` |
| 3 | [$-y$, $x$, 0] | `[11,16]` |
| 4 | [$y$, $-x$, 0] | `[12,15]` |
| 5 | [$-x$, $y$, 0] | `[10,13]` |
| 6 | [$x$, $-y$, 0] | `[9,14]` |
| 7 | [$y$, $x$, 0] | `[4,7]` |
| 8 | [$-y$, $-x$, 0] | `[3,8]` |

Table 6: Wyckoff site: `8e`, site symmetry: `..m`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ x,\ z]$ | `[1,7]` |
| 2 | $[-x,\ -x,\ z]$ | `[2,8]` |
| 3 | $[-x,\ x,\ z]$ | `[11,13]` |
| 4 | $[x,\ -x,\ z]$ | `[12,14]` |
| 5 | $[-x,\ x,\ -z]$ | `[10,16]` |
| 6 | $[x,\ -x,\ -z]$ | `[9,15]` |
| 7 | $[x,\ x,\ -z]$ | `[4,6]` |
| 8 | $[-x,\ -x,\ -z]$ | `[3,5]` |

Table 7: Wyckoff site: `8f`, site symmetry: `.m.`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ z]$ | `[1,14]` |
| 2 | $[-x,\ 0,\ z]$ | `[2,13]` |
| 3 | $[0,\ x,\ z]$ | `[7,11]` |
| 4 | $[0,\ -x,\ z]$ | `[8,12]` |
| 5 | $[-x,\ 0,\ -z]$ | `[5,10]` |
| 6 | $[x,\ 0,\ -z]$ | `[6,9]` |
| 7 | $[0,\ x,\ -z]$ | `[4,16]` |
| 8 | $[0,\ -x,\ -z]$ | `[3,15]` |

Table 8: Wyckoff site: `16g`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[-x,\ -y,\ z]$ | `[2]` |
| 3 | $[-y,\ x,\ z]$ | `[11]` |
| 4 | $[y,\ -x,\ z]$ | `[12]` |
| 5 | $[-x,\ y,\ -z]$ | `[10]` |
| 6 | $[x,\ -y,\ -z]$ | `[9]` |
| 7 | $[y,\ x,\ -z]$ | `[4]` |
| 8 | $[-y,\ -x,\ -z]$ | `[3]` |
| 9 | $[-x,\ -y,\ -z]$ | `[5]` |
| 10 | $[x,\ y,\ -z]$ | `[6]` |
| 11 | $[y,\ -x,\ -z]$ | `[15]` |
| 12 | $[-y,\ x,\ -z]$ | `[16]` |
| 13 | $[x,\ -y,\ z]$ | `[14]` |
| 14 | $[-x,\ y,\ z]$ | `[13]` |
| 15 | $[-y,\ -x,\ z]$ | `[8]` |
| 16 | $[y,\ x,\ z]$ | `[7]` |